

Total No. of Questions :10]

SEAT No. :

P3679

[Total No. of Pages : 3

[5561]-313

B.E. (Information Technology)

DISTRIBUTED SYSTEMS

(2012 Course) (Semester - II) (414461)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Figures to right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain in brief, the key techniques used for indirect communication in Distributed system. [5]
- b) What are various forms of transparency in distributed system? Illustrate network transparency with an example. [5]

OR

- Q2)** a) What is CORBA CDR? Explain marshalling in CORBA along with data structure description in the message using IDL. [5]
- b) What is Group Communication? Discuss IP multicast with respect to following: [5]
- i) Multicast Routers
 - ii) Multicast address allocation

- Q3)** a) Describe how connectionless communication between a client and a server proceeds when using sockets. Explain why are sockets insufficient for high performance communication? [5]
- b) Describe Skype overlay architecture with neat diagram. [5]

OR

- Q4)** a) What is Java Object Serialization and Deserialization? Can sockets and file descriptors be serialized? Explain marshalling a proxy in Java RMI. [5]
- b) Explain the RPC exchange protocols used for implementing various types of RPC. [5]

P.T.O.

- Q5) a)** Two processes P and Q are connected in a ring using two channels, and they constantly rotate a message m. At any one time, there is only one copy of m in the system. Each process's state consists of the number of times it has received m and P sends m first. At a certain point, P has the message and its state is 101. Immediately, after sending m, P initiates the snapshot algorithm. Explain the operation of the algorithm in this case, giving the possible global state (s) reported by it. [8]
- b)** Explain the central server algorithm and evaluate the performance of this algorithm for mutual exclusion. [8]

OR

- Q6) a)** What is Lamport's definition of concurrent events using : [8]
- i) Happened before relation and
 - ii) The concept of causality?

What are the limitations of Lamport's Logical clock?

- b)** Illustrate with suitable example, a 'snapshot' algorithm for determining global states of distributed systems. [8]

- Q7) a)** Discuss Domain Name System with its components like DNS queries, DNS name server. [8]
- b)** Discuss how resource scheduling differs in multimedia applications compared to traditional CPU schedulers. Explain in brief, fair scheduling and real time scheduling methods used for resource scheduling. [8]

OR

- Q8) a)** Explain the principal design feature of BitTorrent file sharing application, its terminology and explain in detail how BitTorrent protocol operates. [8]
- b)** Outline the architecture of X.500 application level directory service standard. Discuss the reasons behind the popularity of LDAP specification over X.500? [8]

Q9) a) What is Protection Domain? Explain the two alternative implementations which are commonly used in distributed systems.

[10]

b) Explain in brief, the use of Cryptography in the implementation of : [8]

i) Secrecy and integrity

ii) Authentication

iii) Digital Signature

OR

Q10)a) Explain in detail the process architecture of KERBEROS with diagram.

[10]

b) Discuss the concept of GRID. Explain in brief, the requirements of Grid Applications. Distinguish between Grid and Cloud Computing. [8]

